

Viktor Trilar

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EDUCATION

Bachelor of Engineering — Data Engineering & AI

Aug 2024 – Present

Turku University of Applied Sciences | GPA 4.4 / 5.00

EXPERIENCE

Research Exchange

Apr – Jul 2026

National Institute of Technology, Sendai College — Japan | Supervisor: Prof. Jun Suzuki

- Built an automated system to verify student breadboard wiring from a single phone photo, replacing manual teacher inspection; DQN wire tracer at 99.4% accuracy and YOLOv8 component detection at 99.5% mAP
- Delivered a bilingual JA/EN web application for instructors to create exercises, receive student submissions, and view class-wide analytics

Applied AI Intern

Feb 2026 – Present

Cognitive Technologies Research Group (CoRe) — Turku, Finland

- Building AI systems for industrial machine failure prediction and 3D part inspection; running large-scale experiments on Finland's national supercomputer (CSC Puhti)

PROJECTS

Predictive Maintenance — Sensor Anomaly Detection

2026

PyTorch · Transformers · Scikit-Learn · CSC Puhti HPC

- Transformer model detecting early machine failure signs from sensor data without historical failure examples; AUC 0.897 across 27 unseen machine types with a 6-model anomaly detection ensemble

3D Part Inspection Pipeline

2026

Python · PyTorch · OpenCV · Open3D · COLMAP

- Takes a video of a manufactured part on a turntable, reconstructs its 3D shape, and compares it against a reference CAD model to find dimensional deviations; sub-1mm accuracy from a phone camera

Maritime Vessel Trajectory Prediction

2026

PyTorch · LSTM · FastAPI · PostgreSQL · MQTT

- Ingests live ship position data from Finland's maritime API and predicts each vessel's location 3, 6, and 12 minutes ahead using an LSTM; served via FastAPI with a live Leaflet.js map dashboard

SKILLS

Programming Python, R

Libraries PyTorch, TensorFlow, Pandas, NumPy, Scikit-Learn, OpenCV

AI / ML Deep learning, transformers, LSTMs, CNNs, anomaly detection, time series

Computer Vision Object detection (YOLOv8), 3D reconstruction (COLMAP, Open3D), Blender

Data Engineering PostgreSQL, FastAPI, Kafka, Airflow, Snowflake, MQTT

Cloud & HPC AWS, Microsoft Azure, SLURM, CSC

LANGUAGES

Slovene Native

English C2 Proficient

German B2 Upper-Intermediate

Finnish B1 Intermediate